Data Specification

Data Specification Document

FireVigil — AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data

SIH Problem Statement ID	SIH26162
Theme	Disaster Management
Sponsoring Organization	National Technical Research Organisation (NTRO)
Official Dataset Link	firms.modaps.eosdis.nasa.gov/map
Document version	1.0
Companion documents	PRD_SIH26162_FireVigil, Architecture_SIH26162_FireVigil

1. Purpose

This document specifies every dataset the system consumes or produces: source, access method, refresh cadence, field-level schema, units, and validation rules. It is the single reference for anyone implementing ingestion, the classification service, or the database layer, and for judges/reviewers who want to verify data provenance against the official PS dataset link.

2. External Data Sources

2.1 NASA FIRMS (Fire Information for Resource Management System) — *Official Dataset*

Official link (as given in PS)	firms.modaps.eosdis.nasa.gov/map
Programmatic access	FIRMS Area/Country API — `https://firms.modaps.eosdis.nasa.gov/api/area/csv/{MAP_KEY}/{SOURCE}/{AREA_COORDINATES}/{DAY_RANGE}`
Auth	Free `MAP_KEY` (API key), registered at firms.modaps.eosdis.nasa.gov
Formats available	CSV, JSON, SHP, KML, WMS
Update cadence	Near-real-time (NRT): new data within ~3 hours of satellite overpass, several passes/day per sensor

Coverage used	India bounding box (configurable), full country supported via `SOURCE=VIIRS_SNPP_NRT`, `MODIS_NRT`, etc.
Historical archive	Available back to 2000 (MODIS) / 2012 (VIIRS) for baseline/training data

Sensor sources consumed:

Source code	Sensor	Spatial resolution	Notes
`MODIS_NRT` / `MODIS_SP`	MODIS (Aqua & Terra)	1 km	Long historical baseline, coarser resolution
`VIIRS_SNPP_NRT`	VIIRS (Suomi-NPP)	375 m	Primary source — finer resolution, better for small industrial fires
`VIIRS_NOAA20_NRT`	VIIRS (NOAA-20)	375 m	Additional daily pass
`VIIRS_NOAA21_NRT`	VIIRS (NOAA-21)	375 m	Additional daily pass

Raw record field schema (CSV/API response):

Field	Type	Unit / Range	Description
`latitude`	float	−90 to 90	Detection center latitude
`longitude`	float	−180 to 180	Detection center longitude
`bright_ti4` / `brightness`	float	Kelvin	Brightness temperature, I-4/Channel 21/22 (fire-sensitive channel)
`bright_ti5`	float	Kelvin	Brightness temperature, I-5 (VIIRS only)
`scan`, `track`	float	km	Pixel size (along scan/track) — used to estimate detection footprint
`acq_date`	date	`YYYY-MM-DD`	Acquisition date
`acq_time`	string	`HHMM` (UTC)	Acquisition time
`satellite`	string	e.g. `N`, `1`, `T`, `A`	Satellite identifier
`instrument`	string	`MODIS` | `VIIRS`	Sensor instrument
`confidence`	string/int	`low` / `nominal` / `high` (VIIRS) or 0–100 (MODIS)	Detection confidence
`version`	string	e.g. `2.0NRT`	Product/collection version
`frp`	float	MW (megawatts)	Fire Radiative Power — proxy for thermal intensity, key classification feature
`daynight`	string	`D` | `N`	Day or night detection
`type` (MODIS only)	int	0–3	0 = presumed vegetation fire, 1 = active volcano, 2 = other static land source, 3 = offshore

> **Classification relevance:** `frp`, `confidence`, `daynight`, `bright_ti4/ti5`, and detection **recurrence at the same coordinates over time** are the primary engineered features feeding the classifier (see Section 4.3).

2.2 OpenStreetMap (OSM) — Industrial Infrastructure Layer

Access method	Overpass API (`https://overpass-api.de/api/interpreter`)
Format	OSM XML / GeoJSON (via `osmtogeojson`)
Refresh cadence	Weekly/on-demand sync (facility locations change slowly)
License	ODbL (Open Database License) — attribution required

Overpass tags queried (mapped to the PS-named facility types):

PS-named facility type	OSM tag(s) queried
Oil refineries	`man_made=works` + `industrial=oil` / `landuse=industrial` + name matching
Petrochemical complexes	`industrial=oil / industrial=chemical`
Thermal power plants	`power=plant` + `plant:source=coal|gas|oil`
Steel industries	`industrial=iron_and_steel` / `landuse=industrial` + name matching
Mining areas	`landuse=quarry`, `man_made=mineshaft`, `industrial=mine`
LNG terminals	`man_made=storage_tank` + `content=lng`, `industrial=gas`

Facility record schema (post-processing into internal DB):

Field	Type	Description
`osm_id`	string	Source OSM node/way/relation ID
`name`	string	Facility name (may be null in OSM — flagged for manual enrichment)
`facility_type`	enum	`refinery` | `petrochemical` | `power_plant` | `steel` | `mining` | `lng_terminal` | `other_industrial`
`geometry`	PostGIS geometry (Point/Polygon)	Facility footprint or centroid
`state`, `district`	string	Derived via reverse geocoding against admin boundaries
`source`	string	`osm` (extendable to future registries, e.g. CPCB)
`last_synced_at`	timestamp	Last successful OSM sync time

2.3 Land-Cover Dataset — *Industrial vs. Natural Discriminator*

Required to satisfy the official deliverable "classification and segregation of industrial fires from forest fires and other natural fires" — land cover at the detection point is a primary discriminating feature.

Candidate sources	ESA WorldCover 10m (2021), Bhuvan/ISRO LULC (national, India-specific), Copernicus Global Land Cover 100m
Format	GeoTIFF raster
Access	Free download (ESA WorldCover via Copernicus/AWS Open Data; Bhuvan via ISRO NRSC portal)
Refresh cadence	Static per release (annual product); re-downloaded on major version updates only

Land-cover classes used for feature engineering:

Class	Relevance
Forest / tree cover	Strong indicator of natural/forest fire when detection falls here and is far from any facility
Cropland	Indicator of agricultural burning
Built-up / industrial	Strong indicator of industrial fire when combined with facility proximity
Bare / sparse vegetation	Indicator of mining activity thermal signature
Grassland / shrubland	Weak indicator, natural fire likely

2.4 Satellite Imagery (Stretch Goal — Visual Confirmation)

Source	Copernicus Data Space Ecosystem — Sentinel-2 (10–20m, optical) and Sentinel-3 SLSTR (thermal)
Access	Sentinel Hub API / openEO, free tier
Format	GeoTIFF / PNG tile, fetched per-event at hotspot coordinates and nearest acquisition date
Usage	Rendered as a thumbnail in the event detail panel for analyst visual confirmation; not used as a primary ML input in the MVP

2.5 Administrative Boundaries

Source	Survey of India / datameet open GeoJSON (state & district boundaries)
Format	GeoJSON
Usage	Region filtering in UI, reverse-geocoding facility/hotspot `state` / `district` fields

---

3. Internal Data Schema (System of Record)

All internal storage in **PostgreSQL 16 + PostGIS**. Geometry columns use SRID 4326 (WGS84).

3.1 `users`

Column	Type	Constraints
`id`	UUID	PK

`name`	text	not null
`email`	text	unique, not null
`password_hash`	text	not null
`role`	enum(`admin` , `analyst` , `viewer`)	default `viewer`
`created_at`	timestamptz	default now()

3.2 `facilities`

Column	Type	Constraints
`id`	UUID	PK
`osm_id`	text	unique per source
`name`	text	nullable
`facility_type`	enum (see §2.2)	not null
`geometry`	geometry(Point/Polygon, 4326)	not null, GiST-indexed
`state`	text	
`district`	text	
`source`	text	default `osm`
`last_synced_at`	timestamptz	

3.3 `hotspots` (raw FIRMS ingestion)

Column	Type	Constraints
`id`	UUID	PK
`latitude`	double precision	not null
`longitude`	double precision	not null
`geometry`	geometry(Point, 4326)	generated from lat/long, GiST-indexed
`acq_date`	date	not null
`acq_time`	text	`HHMM` UTC
`satellite`	text	
`instrument`	enum(`MODIS` , `VIIRS`)	not null
`confidence`	text	raw value from FIRMS
`frp`	numeric	MW, nullable
`bright_ti4`	numeric	Kelvin, nullable
`daynight`	enum(`D` , `N`)	
`raw_payload`	jsonb	full original FIRMS record, for auditability
`ingested_at`	timestamptz	default now()
Constraint	unique(`latitude` , `longitude` , `acq_date` , `acq_time` , `instrument` , `satellite`)	prevents duplicate ingestion

3.4 `classified_events`

Column	Type	Constraints
`id`	UUID	PK
`hotspot_id`	UUID	FK → hotspots, not null
`facility_id`	UUID	FK → facilities, nullable
`primary_class`	enum(`industrial` , `natural`)	not null — **official deliverable #1**
`sub_class`	enum(`industrial_fire` , `gas_flare` , `agricultural_burning` , `mining_activity` , `forest_fire` , `other_natural`)	not null
`land_cover_type`	enum (see §2.3)	
`distance_to_facility_m`	numeric	meters, nullable
`confidence_score`	numeric(0–1)	model output confidence
`model_version`	text	for reproducibility/audit

`is_anomalous`	boolean	true if deviates from facility's historical baseline
`created_at`	timestamptz	default now()

3.5 `facility_baselines`

Column	Type	Description
`id`	UUID	PK
`facility_id`	UUID	FK → facilities
`avg_daily_detections`	numeric	rolling average, detections/day
`avg_frp`	numeric	rolling average FRP (MW)
`std_dev_frp`	numeric	for anomaly z-score computation
`window_start` , `window_end`	date	baseline computation window (e.g., trailing 90 days)
`updated_at`	timestamptz	

3.6 `alerts`

Column	Type	Description
`id`	UUID	PK
`classified_event_id`	UUID	FK → classified_events
`severity`	enum(`high` , `medium` , `low`)	derived from class + anomaly flag
`status`	enum(`new` , `acknowledged` , `resolved` , `false_positive`)	default `new`
`sent_at`	timestamptz	
`acknowledged_by`	UUID	FK → users, nullable

3.7 `feedback`

Column	Type	Description
`id`	UUID	PK
`classified_event_id`	UUID	FK → classified_events
`user_id`	UUID	FK → users
`corrected_label`	text	analyst-supplied correction (feeds future retraining)
`notes`	text	
`created_at`	timestamptz	

---

4. Data Flow & Processing Rules

4.1 Ingestion cadence

Job	Frequency	Source
FIRMS poll	Every 30–60 minutes	FIRMS Area API
OSM facility sync	Weekly	Overpass API
Land-cover raster refresh	On dataset version release (manual trigger)	WorldCover/Bhuvan
Facility baseline recompute	Nightly	Internal, from `hotspots` / `classified_events`

4.2 Deduplication rule

A raw hotspot is considered a duplicate (and skipped) if `(latitude, longitude, acq_date, acq_time, instrument, satellite)` already exists — coordinates are rounded to 5 decimal places (~1.1 m) before comparison to absorb floating-point noise from repeated FIRMS pulls.

4.3 Feature engineering for classification

Feature	Derived from	Purpose
`distance_to_nearest_facility_m`	hotspot geometry ↔ `facilities.geometry` (PostGIS `ST_Distance`)	Core industrial-attribution signal
`facility_type` (if matched)	`facilities.facility_type`	Sector-specific prior (e.g., LNG terminals → higher flare likelihood)

`land_cover_class`	raster lookup at hotspot lat/long	Core natural-vs-industrial discriminator
`recurrence_count_90d`	count of prior hotspots within ~500m over trailing 90 days	Distinguishes persistent source (flare/kiln) from one-off fire
`frp`, `bright_ti4/ti5`	raw FIRMS fields	Thermal intensity signal
`daynight`	raw FIRMS field	Some sources (flares) are highly consistent day/night; wildfires vary
`z_score_frp`	`(frp − facility_baselines.avg_frp) / std_dev_frp`	Anomaly detection — flags fires that exceed a facility's "normal" thermal signature

4.4 Classification decision logic (summary)

1. If `land_cover_class` = forest/grassland/cropland **and** `distance_to_nearest_facility_m` > threshold (e.g., 1 km) → `primary_class = natural`, `sub_class` = `forest_fire` or `agricultural_burning` (by land cover).
2. If matched to a facility within threshold **and** `recurrence_count_90d` is high with low `z_score_frp` → `primary_class = industrial`, `sub_class = gas_flare` (persistent, expected).
3. If matched to a facility **and** `z_score_frp` is high (anomalous spike) or `recurrence_count_90d` is low (new/rare) → `primary_class = industrial`, `sub_class = industrial_fire`, `is_anomalous = true` → triggers high-severity alert.
4. If matched to a mining-tagged facility/land cover → `sub_class = mining_activity`.
5. Ambiguous/low-confidence cases are queued for analyst review via the feedback loop (§3.7).

4.5 Data retention

Dataset	Retention
Raw `hotspots`	Indefinite (partitioned by month for query performance)
`classified_events`	Indefinite (linked to raw hotspot for audit trail)
`raw_payload` (jsonb)	Indefinite — preserves original FIRMS record for reproducibility
Satellite imagery thumbnails	12 months rolling (object storage lifecycle rule)

---

5. Data Quality & Validation Rules

- Reject/flag hotspot records with `confidence = low` **and** no facility match **and** no land-cover data available (insufficient signal to classify reliably) — routed to a `sub_class = unclassified` bucket rather than forced into a guess.
- `latitude` / `longitude` must fall within the configured area-of-interest bounding box; out-of-range records are dropped at ingestion.
- Facility geometries with `name IS NULL` are retained but flagged `needs_review` for manual name enrichment (OSM data quality varies by region).
- All classification outputs must carry `model_version` for reproducibility and to support before/after comparison when the model is retrained on accumulated `feedback`.

---

6. Data Provenance & Compliance

- **NASA FIRMS**: Data provided by LANCE, part of NASA's Earth Observing System Data and Information System (EOSDIS). Any published output or shared imagery must include the required FIRMS attribution/disclaimer per NASA's data use guidelines.
- **OpenStreetMap**: ODbL license — attribution ("© OpenStreetMap contributors") required on all map views.
- **Land-cover datasets**: Attribution per source license (ESA WorldCover — CC BY 4.0; Bhuvan — per ISRO/NRSC terms).
- No personally identifiable information (PII) is collected from any external dataset; internal `users` table PII (name, email) is limited to system operators/analysts and is not derived from any external source.